FACULTY OF COMPUTING
UTM Johor Bahru

SCST1123: MATHEMATICS FOR SOFTWARE ENGINEERING
20242025/SEMESTER 1

ASSIGNMENT #3a (Chapter 5)

INSTRUCTION:

1. This is an assignment for a **GROUP of 3 students**. Please clearly write the course details including the group members name & matric number in the front page of the submission.
2. This assignment will give **5%** of overall course marks.
3. The submission is must be neatly written using handwriting. Any submission with handwriting that is unreadable, will be **REJECTED**.
4. Due date: **Monday - 23/12/2024, 12.00 pm**.

Question 1 **[4 Marks]**

Using the digits 1, 2, 3, 4 and 5, how many 4 digit numbers can be formed if:

a) The first digit must be 1 and repetition of the digits is allowed? (1 mark)
b) The first digit must be 1 and repetition of the digits is not allowed? (1 mark)
c) The number must be divisible by 2 and repetition is allowed? (2 marks)

Question 2 **[6 Marks]**

A father, mother, 2 boys, and 3 girls are asked to line up for a photograph. Determine the number of ways they can line up if:

a) there are no restrictions. (2 marks)
b) the parents stand together. (2 marks)
c) all the females stand together. (2 marks)

Question 3 **[6 Marks]**

How many words, with or without meaning can be made from letters of the word TUESDAY, assuming that no letter is repeated, if:

a) 4 letters are used at a time. (2 marks)
b) All letters are used at a time. (2 marks)
c) All letters are used but first letter is a vowel. (2 marks)

Question 4 **[9 Marks]**

A committee of 7 must be formed from 9 boys and 4 girls. In how many ways this can be done when the committee consist of:

a) Exactly 3 girls. (2 marks)

b) At least 3 girls. (3 marks)

c) At most 3 girls. (4 marks)

Question 5 **[7 Marks]**

A bracelet is made from 5 alphabet beads as shown in Figure 1.

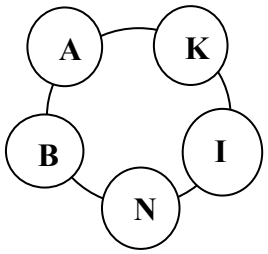


Figure 1

a) How many designs of the bracelet can be made from the 5 beads in any possible arrangement? (1 mark)

b) How many designs of the bracelet can be made if the bead with alphabet **A** and **N** should be side by side? (1 mark)

c) How many designs of the bracelet can be made if the bracelet should consists the sequence of beads **A**, **I**, **N** in any arrangement? (3 marks)

d) If the 5 beads are cut loose from the bracelet and rearranged in linear sequence, how many initials of 3 alphabet beads can be made? (2 marks)